

RATINGS

MODEL	STANDBY RATING	PRIME RATING	SPEED	POWER FACTOR
WM-220 KVA	220 kVA 176 kW	200 kVA 160 kW	1500 rpm	0.8 Cos Phi

STANDARD VOLTAGE	STANDBY KVA	STANDBY KW	PRIME KVA	PRIME KW
400 / 230 V	220	176	200	160

DESCRIPTION

WM-220 KVA diesel generator set: water-cooled three-phase 50 hz diesel, 1500 rpm, 400 / 230 V, power factor 0.8. Engine WIRMAN W-022VW, alternator WIRMAN SF274G, control unit DSE6120 MKIII.

WIRMAN has ISO 9001, ISO 14001 and OHSAS 18001 quality certifications.

STANDARD GENERATOR FEATURES

- Heavy duty water-cooled diesel engine and brushless alternator
- Solid steel base frame with minimum 12 hours base tank and anti-vibration mountings
- AMF control panel with digital automatic main control module
- 50 °C tropical radiator
- Battery charge rectifier, lead-acid battery
- Residential type silencer
- Flexible exhaust compensator
- Engine block water heater unit
- All rotating parts covered by metal mesh guards
- Engine and alternator manufacturer test reports (if provided)
- Factory load, performance and function tests
- User manual, electrical drawing and guarantee certificate

RATINGS DEFINITIONS

PRIME POWER (PRP)

Prime power is the maximum power which a generating set is capable of delivering continuously whilst supplying a variable electrical load when operated for an unlimited number of hours per year under the agreed operating conditions with the maintenance intervals and procedures being carried out as prescribed by the manufacturer. The permissible average power output (Ppp) over 24 h of operation shall not exceed 70 % of the PRP.

EMERGENCY STANDBY POWER (ESP)

Emergency standby power is the maximum power available during a variable electrical power sequence, under the stated operating conditions, for which a generating set is capable of delivering in the event of a utility power outage or under test conditions for up to 200 h of operation per year with the maintenance intervals and procedures being carried out as prescribed by the manufacturer. The permissible average power output over 24 h of operation shall not exceed 70 % of the ESP.

WM-220 KVA

PRODUCT PHOTO

SOUNDPROOF CANOPY SPECIFICATIONS

- Special design for minimising acoustic level
- Galvanized steel construction further protected by polyester powder coat paint
- Black finish stainless steel locks and hinges
- Control panel viewing window in a lockable access door
- Emergency stop push button (red) mounted on enclosure exterior
- Lifting, drag and jacking points on base frame
- Radiator fill via removable, flush mounted rain cap fitted with compression seal

CODES AND STANDARDS

WIRMAN has ISO 9001, ISO 14001 and OHSAS 18001 quality certifications. WIRMAN gensets are CE marked in accordance with the following directives:

- 2006/42/EC Machinery safety
- 2014/30/EU Electromagnetic compatibility
- 2014/35/EU Electrical equipment designed for use within certain voltage limits
- 2000/14/EC Sound power level, noise emissions of outdoor equipment (amended by 2005/88/EC)
- 97/68/EC Emissions of gaseous and particulate pollutants (amended by 2002/88/EC & 2004/26/EC)
- EN 12100, EN 13857, EN 60204

GENERATOR SET SPECIFICATIONS

Frequency	50	Hz	Power factor	0.8	Cos Phi
Engine speed	1500	rpm	Phases / poles	3 / 4	
Standard voltage	400 / 230	V	Connection type (standard)	Star	

ENGINE SPECIFICATIONS

WIRMAN W-022VW

Rated Output	176	kW	Compression Ratio	16.4:1	
Manufacturer	WIRMAN		Fuel Consumption Standby (110%)	33	l/h
Model	W-022VW		Fuel Consumption 100% PRP	31.4	l/h
Engine Type	4 Stroke - Diesel		Fuel Consumption 75 % PRP	25.2	l/h
Injection Type	Direct Injection		Fuel Consumption 50 % PRP	18.5	l/h
Aspiration Type	Turbo Charged		Lube oil consumption with full load	< 0.5	l/h
Number of cylinders and arrangement	In line		Total oil capacity including tubes, filters	25	L
Bore and Stroke	111 × 139	mm	Total coolant capacity	40	L
Displacement	—	L	Governor	Mechanical	Type
Cooling System	Water + 50 % antifreeze		Air Filter	Dry	Type
Lube Oil Specifications	ACEA E3,E4,E5 (15W-40)				

ALTERNATOR SPECIFICATIONS

WIRMAN SF274G

Manufacturer	WIRMAN		Exciter system	Self-excited, brushless	
Model	SF274G		Voltage regulator	A.V.R (Electronic)	
Poles	4	No.	Bracket type	Single Bearing	
Connection type (standard)	Star		Coupling system	Flexible disc	
Total Harmonic Content	< 2 %		Efficiency (%)	87 %	
Insulation	H Class	Class	Voltage Output (VAC)	230 / 400	
Enclosure (according to IEC 34-5)	IP 21/23		Frequency (Hz)	50	

FUEL CONSUMPTION

SPECIFICATIONS 50 HZ / 1500 RPM

LOAD	STANDBY (110 %)	100 % PRP	75 % PRP	50 % PRP	UNIT
Fuel consumption	33	31.4	25.2	18.5	l/h

Note: All data based on operation to BS 5514 and DIN 6271 standard reference conditions.

CONTROL SYSTEM

DSE6120 MKIII

KEY FEATURES

- 4-line back-lit LCD text display
- Multiple display languages
- Five-key menu navigation
- LCD alarm indication
- Customisable power-up text and screen images
- DSENet® expansion compatibility
- Data logging facility
- Internal PLC editor
- Protections disable feature
- Fully configurable via PC using USB communications
- Front panel configuration with PIN protection
- Power save mode
- 3-phase generator sensing and protection
- 3-phase mains (utility) sensing and protection (DSE6120 MKIII only)
- Automatic load transfer control (DSE6120 MKIII only)
- Generator current and power monitoring (kW, kvar, kVA, pf)
- Mains (utility) current and power monitoring (kW, kvar, kVA, pf) (DSE6120 MKIII only)
- kW overload alarm
- Over current protection
- Breaker control via fascia buttons
- Fuel and start outputs configurable when using CAN
- 6 configurable DC outputs
- 4 configurable analogue/digital inputs
- 8 configurable digital inputs
- CAN, MPU and alternator frequency speed sensing in one variant
- Real time clock
- Manual and automatic fuel pump control
- Engine pre-heat and post-heat functions
- Engine run-time scheduler
- Engine idle control for starting & stopping
- Fuel level alarms
- 3 configurable maintenance alarms
- Compatible with a wide range of CAN engines, including Tier 4 engine support
- Uses DSE Configuration Suite PC Software for simplified configuration
- Licence-free PC software
- IP65 rating (with optional gasket) offers increased resistance to water ingress
- Configurable CAN read & transmitted information
- 1 alternative configuration



- 1 Stop / Reset Mode
- 2 Manual Mode
- 3 Test Mode (DSE6120 MKIII only)
- 4 Auto Mode
- 5 Alarm Mute / Lamp Test
- 6 Start
- 7 Menu Navigation

KEY BENEFITS

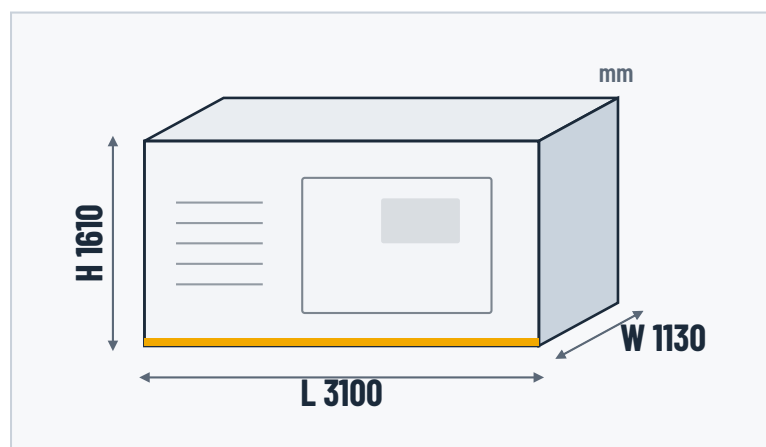
- Automatically transfers between mains (utility) and generator (DSE6120 MKIII only) for convenience
- Hours counter provides accurate information for monitoring and maintenance periods
- User-friendly set-up and button layout for ease of use
- Multiple parameters are monitored & displayed simultaneously for full visibility
- The module can be configured to suit a wide range of applications for user flexibility
- PLC editor allows user configurable functions to meet user specific application requirements

RELATED MATERIALS

- DSE6110 MKIII & DSE6120 MKIII Installation Instructions
- DSE6110 MKIII & DSE6120 MKIII Operator Manual
- DSE6110 MKIII & DSE6120 MKIII Configuration Suite PC Manual

WEIGHTS AND DIMENSIONS

CANOPIED



CANOPIED SET		
Length (L)	3100	mm
Width (W)	1130	mm
Height (H)	1610	mm
Weight	1855	kg
Fuel tank	200	lt